

System One Models: Typed Decisions Instead of Generated Text

Learn when a specialised model beats a general-purpose LLM, and exactly what you give up to get it.

For engineers who reach for a frontier LLM by default and want to know when that is the wrong tool · 22 September 2026

The one- or two-page version: what matters, the topics it covers, and every term defined. Read the full lesson for the explanations and worked examples.

[Watch the source video](#)[Read the full lesson](#)[Read the condensed version](#)

What matters

- ✓ A System One model takes unstructured state and returns a schema-constrained value with a calibrated probability, instead of free-form text.
- ✓ Jev's speed is architectural rather than tuning: it samples every output in one pass, escaping the one-token-after-another floor of autoregressive decoding.
- ✓ TypeSafe's headline figures — 70–500 ms end to end, 40x–200x faster than frontier LLMs — are self-reported; the most grounded public number is Vercel's 18x at p95 on a live safety-review step.
- ✓ The model cannot hallucinate a fact into a sentence because it cannot produce a sentence: every output must fit a schema fixed before the call.
- ✓ RLHF optimises for human preference, RLVR for verifiable reward, RLCD for calibration — the objective a model was trained against is what limits where it fits.
- ✓ Giving up text is a real cost: no explanation, no chain of thought, and reported accuracy near mid-tier models rather than frontier ones.
- ✓ The practical move is a portfolio, not a winner: a generalist for open-ended work, a specialist for the high-volume decisions inside a workflow.

What it covers

- 01 The orthodox path, and what it optimised for** 0:00
Models have been tuned for the two use cases that dominated the application layer — chat, then coding agents — and the layers below reshaped themselves to serve them.
- 02 The gap: quick decisions under uncertainty** 0:40
Workflow automation stayed out of reach not because models were not smart enough, but because they were optimised against objectives that do not transfer to it.
- 03 Why the speed is architectural, not tuning** 2:04
An autoregressive model has a latency floor set by its decoding loop; parallel sampling removes the loop rather than making it faster.
- 04 Typed in, typed out: the rules of engagement change** 4:08
You stop writing prompts and start declaring a schema; the model returns a value inside that schema with a probability attached.
- 05 Primitives you compose, like logic gates** 4:48
The model offers a few decision shapes, and anything complicated is built by composing them in your own code.

06 **Calibration as the training objective** 6:11

RLCD optimises the probabilities against outcomes rather than against what a person prefers, which is what makes the confidence number worth branching on.

07 **What you give up, and what it costs** 5:31

No prose means no explanation, no chain of thought, and — on TypeSafe's own benchmark — accuracy closer to mid-tier models than to the frontier.

08 **Choosing: a portfolio, not a winner** 6:53

Ask what shape the task is before asking which model is best — bounded answers at volume with no human reader is a different job from open-ended generation.

Glossary

System One model — TypeSafe's name for a model class that takes unstructured state and returns a typed, schema-constrained decision with a probability, rather than free-form text.

Parallel sampling — Computing all of a request's outputs in a single pass instead of in sequence, so the number of questions barely affects latency.

RLCD — Reinforcement Learning for Calibrated Decisions — TypeSafe's training method, which optimises the model's probabilities against outcomes rather than human preference.

RLVR — Reinforcement Learning with Verifiable Rewards — training against answers that can be checked automatically, such as a test suite passing.

p95 latency — The response time that 95% of requests come in under — a truer picture of a production path than an average.

Autoregressive decoding — Generating output one token at a time, each conditioned on the tokens before it — the reason a language model's response time grows with its length.

Calibration — The property that a model's stated confidence matches its real accuracy: answers given at 0.9 confidence are right about 90% of the time.

RLHF — Reinforcement Learning from Human Feedback — training a model to produce the answers human raters prefer.

Schema-constrained output — Output whose shape and allowed values are fixed before the call, so a malformed or out-of-range answer cannot be produced.

Pareto frontier — The set of options where improving one dimension, such as latency, requires giving up another, such as generality.

Explanations are AI-generated. Check anything you intend to rely on.